

Pure Mathematical Conal Architecture (PMCA) Generation 6.0: Conal Metric Geometry and Information-Geometric Optimization for Neuro-Symbolic Substrates

Author: Jonathan Wayne Fleuren Affiliation: Independent Researcher — Aetherius
Cognitive Systems, Gatineau, Quebec, Canada E-mail: contact@aetherius.ai Zenodo Record
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ABSTRACT

Auto-regressive Large Language Models (LLMs) are known to exhibit non-deterministic hallucination on closed-form symbolic tasks, quadratic self-attention complexity ($\mathcal{O}(N^2)$), and limited interpretability of their latent representations. We present the Pure Mathematical Conal Architecture (PMCA), a neuro-symbolic cognitive substrate that decouples internal state evolution from natural language generation by treating natural language strictly as a Post-Processing Communicative Translation Layer (PPCTL). Internal state transformations are executed first within a continuous 3D Conal Metric Geometry (z, r, θ) , with a symbolic-verification substrate (SymPy) providing exact, non-hallucinating solutions for closed-form calculus and algebra. PMCA introduces a self-structuring information-geometric manifold whose metric tensor evolves via a Ricci-flow-inspired matrix relaxation rule ($\frac{dg_{ij}}{d\tau} = -2\mathcal{R}_{\text{norm}}g_{ij} + \alpha F_{ij}$), using a curvature-proxy term $\mathcal{R}_{\text{norm}}$ rather than the full Ricci curvature tensor. We detail a shape-stable $(N, 6)$ XLA Geometry Transcription Layer designed to avoid graph recompilation on Google TPU v5e accelerators; an Integrity Verification Protocol enforcing cryptographic SHA-256 state hashing to detect tampering of intermediate geometry states during translation; and a Geometry Discovery Engine that classifies emergent metric signatures using topological fingerprints. We report empirical Cloud TPU v5e measurements in Section 6 and present systematic component ablations isolating the contribution of each architectural layer. We scope our claims to what is directly demonstrated: exact symbolic verification for closed-form mathematics, and a deterministic, inspectable geometric state-tracking layer. We do not claim to resolve hallucination, representational opacity, or persona-alignment failure modes for open-ended natural-language reasoning outside these domains; Section 7 discusses this scope explicitly.

Index Terms: Information geometry, Riemannian metrics, neuro-symbolic AI, symbolic verification, XLA acceleration, shape-stable tensors, model interpretability.

1. INTRODUCTION

Modern artificial intelligence is dominated by token-based auto-regressive Transformer architectures [1]. While statistical text generation achieves impressive surface fluency, this paradigm exhibits four commonly-cited structural limitations:

1. **Epistemic Instability and Hallucination:** Token probability maximization prioritizes textual plausibility over underlying mathematical truth, generating convincing falsehoods without internal verification [2], [3].
2. **Quadratic Compute Complexity ($\mathcal{O}(N^2)$):** Re-evaluating dense self-attention matrices for identical or logically equivalent queries incurs substantial hardware

energy expenditure without state reuse [68].

3. **Black-Box Opacity:** Latent Transformer representations lack interpretable geometric invariants, rendering real-time auditability difficult prior to output decoding [4].
4. **Brittle Persona Alignment:** Forcing human social personas onto statistical token generators creates identity paradoxes, vulnerability to adversarial jailbreaks, and alignment faults [5].

PMCA directly targets the first of these: for closed-form symbolic mathematics, it provides exact, non-hallucinating verification by routing computation through a symbolic solver rather than statistical token generation. It further provides a deterministic, inspectable geometric substrate for state-tracking, whose relationship to the remaining three pathologies (compute complexity, opacity, and persona alignment) is architectural rather than empirically demonstrated in this manuscript — we return to this scope distinction explicitly in Section 7.

PMCA operationalizes the cognitive thesis that reasoning occurs in a non-verbal representational substrate (*Mentalese*) structurally distinct from surface natural language [6]–[8]. Rather than treating text as the cognitive medium, PMCA treats natural language strictly as a Post-Processing Communicative Translation Layer (PPCTL) [9], [10]. All internal inference, state evolution, and physical simulations are executed within a Pure 3D Conal Metric Geometry (z, r, θ) first, with natural language generated subsequently to decode verified mathematical ground truth [11], [12].

By grounding cognition in continuous Riemannian differential geometry [13], [14], non-equilibrium information thermodynamics [15], [16], and hybrid neuro-symbolic solvers [17], [18], PMCA establishes a deterministic, non-anthropomorphic cognitive substrate running natively on XLA hardware accelerators [19].

2. RELATED WORK

2.1 Geometric Deep Learning and Non-Euclidean Embeddings

Geometric Deep Learning (GDL) generalizes neural networks to non-Euclidean domains, including graphs and Riemannian manifolds [13]. Nickel and Kiela [20] demonstrated that Poincaré hyperbolic embeddings capture hierarchical structures far more efficiently than Euclidean spaces. Ganea et al. [22] extended this to hyperbolic neural networks, while Chami et al. [50] developed hyperbolic graph convolutional networks. PMCA builds upon these non-Euclidean paradigms by introducing a dynamic, tapering 3D conal metric geometry (z, r, θ) where longitudinal depth z models optimization convergence, radius $r(z)$ bounds contracting information entropy, and phase angle θ encodes contextual orientation.

2.2 Neuro-Symbolic Integration and Physics-Informed Models

Hybrid neuro-symbolic architectures aim to combine the pattern-recognition capabilities of neural networks with the formal precision of symbolic logic [17]. Physics-Informed Neural Networks (PINNs) [18] incorporate differential equations into loss functions, while Lagrangian and Hamiltonian Neural Networks [46], [51] embed physical conservation laws directly into network dynamics. PMCA advances neuro-symbolic integration by

maintaining a dual-engine operator manifold that pairs SymPy closed-form AST symbolic verification with 3D classical particle physics integration.

2.3 Information Geometry and Metric Flow Optimization

Information geometry applies differential geometry to probability distributions, utilizing the Fisher Information Metric (FIM) as a Riemannian metric tensor [66]. Hamilton [64] and Perelman [65] established Ricci flow ($\frac{\partial g_{ij}}{\partial t} = -2R_{ij}$) as a powerful intrinsic metric deformation process, where R_{ij} is the true Ricci curvature tensor, derived from the manifold's Christoffel connections. In PMCA Generation 6.0, we adopt a **Ricci-flow-inspired** metric relaxation rule that replaces the true Ricci tensor with a cheaply-computable curvature proxy $\mathcal{R}_{\text{norm}}$ (Eq. 13), combined with a data-driven Fisher Information Metric term, allowing the metric to self-structure from data statistics without the cost of computing the full curvature tensor at every step. We are explicit that this is an engineering approximation motivated by Ricci flow, not a claim of mathematical equivalence to Hamilton–Perelman Ricci flow.

2.4 Cognitive Architectures and Latent Energy Models

Classic cognitive architectures such as ACT-R [71], SOAR [19], and LIDA [55] model cognitive processes through unified modular pipelines. LeCun [5], [31] proposed Energy-Based Models (EBMs) and modular joint-embedding architectures to eliminate statistical autoregression. PMCA operationalizes these cognitive and energy-based principles by utilizing a scalar potential drive $V(z)$ to pull state trajectories into focal energy attractors prior to output decoding.

3. MATHEMATICAL AND TOPOLOGICAL FOUNDATIONS

3.1 3D Encased Conal Metric Pathway

The PMCA cognitive manifold is structured as a tapering conical metric space in cylindrical polar coordinates (z, r, θ) , where $z \in [0, Z_{\text{max}}]$ represents longitudinal depth, $r(z) = r_0(1 - \alpha \frac{z}{Z_{\text{max}}})$ defines the contracting radius ($\alpha \in (0, 1)$), and $\theta \in [0, 2\pi)$ denotes circumferential phase angle [21]. State tensors $\mathbf{X} \in \mathbb{R}^{n \times d}$ undergo spatial metric scaling governed by a position-dependent Riemannian tensor field [23]:

$$\mathcal{M}_{\text{conal}}(z, r, \theta) = \mathbf{X} \cdot \mathbf{g}(z, r, \theta), \quad (1)$$

where the regularized metric-scaling matrix $\mathbf{g}(z, r, \theta)$ is defined as:

$$\mathbf{g}(z, r, \theta) = \text{diag} \left(1 + \frac{z}{Z_{\text{max}}}, \frac{r(z)}{r_0}, 1 + 0.1 \cos \theta \right). \quad (2)$$

Theorem 1 (Positive-Definiteness and Non-Degeneracy of $\mathbf{g}$).

*For any conal depth $z \in [0, Z_{\text{max}}]$, radius $r(z) \in [r_0(1 - \alpha), r_0]$, and polar phase angle $\theta \in [0, 2\pi)$, the metric scaling tensor field $\mathbf{g}(z, r, \theta)$ defined in (2) is strictly positive-definite ($\mathbf{g} \succ 0$) and non-degenerate ($\det(\mathbf{g}) > 0$).

Proof: Matrix $\mathbf{g}(z, r, \theta)$ is diagonal with eigenvalues $\lambda_1 = 1 + \frac{z}{Z_{\max}}$, $\lambda_2 = \frac{r(z)}{r_0}$, and $\lambda_3 = 1 + 0.1 \cos \theta$. Since $z \geq 0$ and $Z_{\max} > 0$, $\lambda_1 \geq 1.0 > 0$. Since $r(z) = r_0(1 - \alpha \frac{z}{Z_{\max}})$ with $\alpha \in (0, 1)$ and $z \in [0, Z_{\max}]$ closed, $\lambda_2 \in [1 - \alpha, 1] > 0$ (both endpoints are attained: $\lambda_2 = 1$ at $z = 0$, $\lambda_2 = 1 - \alpha$ at $z = Z_{\max}$). Since $\cos \theta \in [-1, 1]$, $\lambda_3 = 1 + 0.1 \cos \theta \in [0.9, 1.1] > 0$. Thus $\det(\mathbf{g}) = \lambda_1 \lambda_2 \lambda_3 \geq (1.0)(1 - \alpha)(0.9) > 0$, guaranteeing non-degeneracy and eliminating coordinate singularities in the inverse tensor g^{ij} . ■

3.2 External 3D Encasing Lattice and Vector Field Induction

An outer 3D vector lattice ($r_{\text{outer}} = r(z) + \Delta r$) encases the inner conal pathway [25]. The casing emits inward induction forces $\mathbf{F}_{\text{ext}}$ acting on interior state vectors $\mathbf{p}_i$:

$$\mathbf{F}_{\text{ext}}(\mathbf{p}_i) = \sum_{k \in \text{Outer Shell}} \mathbf{E}_k(\mathbf{r}_k), \quad \mathbf{E}_k(\mathbf{r}) = -\frac{\mathbf{r}}{\|\mathbf{r}\|^2 + \epsilon}, \quad \mathbf{r}_k = \mathbf{p}_i - \mathbf{v}_k. \quad (3)$$

Lemma 2 (Local Divergence of a Single Casing Vertex).

For a single casing vertex $\mathbf{v}_k$ and displacement $\mathbf{r} = \mathbf{p}_i - \mathbf{v}_k$, the divergence of the regularized field $\mathbf{E}_k(\mathbf{r})$ is:

$$\nabla \cdot \mathbf{E}_k(\mathbf{r}) = -\frac{\|\mathbf{r}\|^2 + 3\epsilon}{(\|\mathbf{r}\|^2 + \epsilon)^2}, \quad (4)$$

*which is strictly negative for every $\mathbf{r}$ and every $\epsilon > 0$, and approaches $-\frac{1}{\|\mathbf{r}\|^2}$ as $\epsilon \rightarrow 0^+$.

Each casing vertex is therefore individually a local sink; no single vertex's field is divergence-free.*

Proof: Direct differentiation of component $E_x = \frac{-x}{x^2 + y^2 + z^2 + \epsilon}$ yields $\partial_x E_x = -\frac{(\|\mathbf{r}\|^2 + \epsilon) - 2x^2}{(\|\mathbf{r}\|^2 + \epsilon)^2}$.

Summing partials over three spatial dimensions yields Equation (4). Since $\|\mathbf{r}\|^2 \geq 0$ and $\epsilon > 0$, both numerator and denominator of the unsigned ratio are strictly positive, so $\nabla \cdot \mathbf{E}_k < 0$ for all $\mathbf{r}$. As $\epsilon \rightarrow 0^+$, this tends to $-1/\|\mathbf{r}\|^2$. ■

Theorem 2 (Antipodal Force-Vector Cancellation at the Geometric Center).

*For a symmetric 3D lattice shell in which every vertex $\mathbf{v}_k$ has an antipodal partner $\mathbf{v}'_k = -\mathbf{v}_k$ (relative to the shell's center, taken as the origin), the net encasing force **vector** evaluated exactly at the center vanishes:*

$$\mathbf{F}_{\text{ext}}(\mathbf{0}) = \sum_{k=1}^{N/2} (\mathbf{E}_k(\mathbf{0}) + \mathbf{E}'_k(\mathbf{0})) = \mathbf{0}. \quad (4a)$$

Proof: At $\mathbf{p}_i = \mathbf{0}$, $\mathbf{r}_k = -\mathbf{v}_k$, so $\mathbf{E}_k(\mathbf{0}) = \frac{\mathbf{v}_k}{\|\mathbf{v}_k\|^2 + \epsilon}$. For the antipodal partner $\mathbf{v}'_k = -\mathbf{v}_k$:

$\mathbf{E}'_k(\mathbf{0}) = \frac{\mathbf{v}'_k}{\|\mathbf{v}'_k\|^2 + \epsilon} = \frac{-\mathbf{v}_k}{\|\mathbf{v}_k\|^2 + \epsilon} = -\mathbf{E}_k(\mathbf{0})$. Each antipodal pair therefore contributes

$\mathbf{E}_k(\mathbf{0}) + \mathbf{E}'_k(\mathbf{0}) = \mathbf{0}$ exactly, and summing over all $N/2$ pairs gives $\mathbf{F}_{\text{ext}}(\mathbf{0}) = \mathbf{0}$. ■

Remark (on the distinction between Lemma 2 and Theorem 2). These are two different mathematical claims about two different objects, and it is important not to conflate them. Lemma 2 is a statement about the scalar divergence of a single field — a quantity that is

strictly negative everywhere by construction and *cannot* be driven to zero by symmetric placement of vertices, since divergence has no direction for antipodal terms to cancel against. Theorem 2 is a statement about the vector sum of forces evaluated at one specific point (the center) — a quantity that genuinely does vanish by symmetry, since force vectors from antipodal vertices point in exactly opposite directions at that point. The empirically-measured "zero net divergence" figure reported in Table 1 (Section 6.2) refers to the divergence of the fully-integrated simulated particle velocity field — which incorporates the gravitational, spring, and drag dynamics of Path B (Section 5.1) in addition to the casing force — and should be understood as a property of that composite simulated field, not of the closed-form casing field $\mathbf{E}_k$ in isolation, which by Lemma 2 can never have zero divergence at any point.

3.3 Perceptual Input Interface and Phase-Space Feature Extraction

To initialize state representations without embedding continuous reasoning inside natural language models, PMCA treats dense text encodings strictly as an external perceptual boundary operator $\mathcal{E}_{\text{in}} : \mathcal{X}_{\text{text}} \rightarrow \mathbb{R}^D$ ($D = 384$) using a frozen all-MiniLM-L6-v2 encoder [73]. Internal cognitive dynamics, optimization, and state evolution occur strictly downstream within the conal metric space $\mathcal{M}_{\text{conal}}(z, r, \theta)$.

The singular value spectrum $\mathbf{S} = (\sigma_1, \sigma_2, \dots, \sigma_D)$ of the normalized embedding outer product $\mathbf{A} = \mathbf{e}\mathbf{e}^T$ establishes the initial Shannon spectral entropy:

$$H_{\text{ling}} = - \sum_{i=1}^D \hat{\sigma}_i \log_2(\hat{\sigma}_i + \epsilon), \quad \text{where } \hat{\sigma}_i = \frac{\sigma_i}{\sum_j \sigma_j}. \quad (5)$$

To determine the initial conal entrance phase angle $\theta_{\text{affective}} \in [0, 2\pi)$, the high-dimensional vector $\mathbf{e}$ is projected onto two orthonormal reference vectors $\mathbf{u}_v, \mathbf{u}_a \in \mathbb{R}^D$ establishing affective valence and arousal axes, in the sense of Russell's circumplex model of affect [69] (derived via offline singular value decomposition of embeddings from the NRC Valence-Arousal-Dominance Lexicon [72], fixed during deployment):

$$x_v = \tanh(\mathbf{e} \cdot \mathbf{u}_v), \quad y_a = \tanh(\mathbf{e} \cdot \mathbf{u}_a), \quad \theta_{\text{affective}} = \text{atan2}(y_a, x_v) \pmod{2\pi}. \quad (6)$$

The 2D affective pair (x_v, y_a) directly parameterizes the initial state vector at the conal entrance ($z = 0$):

$$\mathbf{p}_0 = (z_0 = 0, r_0 = r(0), \theta_0 = \theta_{\text{affective}}) \in \mathcal{M}_{\text{conal}}, \quad (7)$$

establishing a deterministic linear projection from perceptual embedding space into the 3D conal phase space.

3.4 Gradient Potential Drive $V(z)$ and Analytical Force

State vector acceleration toward ground-truth solutions is driven by a scalar potential field $V(z)$ along the longitudinal z -axis [31], [32]:

$$V(z) = - \frac{k_{\text{focal}}}{(Z_{\text{max}} - z + \epsilon)^2} + \gamma \cdot s_{\text{entropy}} \cdot e^{-\lambda z}. \quad (8)$$

The analytical driving force $\mathbf{F}_{\text{drive}}(z) = -\nabla V(z)$ is derived as [33], [34]:

$$\mathbf{F}_{\text{drive}}(z) = \left(\frac{2k_{\text{focal}}}{(Z_{\text{max}} - z + \epsilon)^3} + \lambda \gamma s_{\text{entropy}} e^{-\lambda z} \right) \hat{\mathbf{z}}. \quad (9)$$

At the conal entrance ($z \approx 0$), $e^{-\lambda z}$ provides an exponential push away from unstructured disorder [35]. Near the apex ($z \rightarrow Z_{\text{max}}$), the non-linear denominator surges, pulling state vectors into singular focal energy attractors [36], [37].

3.5 PMCA Entropy Dissipation Law

Thermodynamic coherence across multi-scale fractal sub-cones is enforced by a subadditivity constraint and a critical entropy governor [15], [16], [38]:

$$H_{\text{parent}} \leq \sum_i H_{\text{child}}^{(i)}, \quad \text{with equality iff } \langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0. \quad (10)$$

Theorem 3 (Entropy Subadditivity Theorem [67]).

Let $V = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$ be a set of state vectors in a fractal sub-cone, and let $\mathbf{K}$ denote their normalized Gram matrix. Define the joint Gram entropy $H_{\text{Gram}}(V) \equiv \log(\det \mathbf{K})$ — a log-determinant entropy associated with the covariance structure of V , distinct from the per-vector spectral entropy H_{ling} of Eq. (5). Then $H_{\text{Gram}}(V) \leq \sum_{i=1}^m \log(K_{ii})$, with equality holding if and only if all state vectors are mutually orthogonal.

Proof: By Hadamard's inequality [67], $\det(\mathbf{K}) \leq \prod_{i=1}^m K_{ii}$ with equality iff $\mathbf{K}$ is diagonal. Taking logarithms yields $\log(\det \mathbf{K}) \leq \sum_{i=1}^m \log(K_{ii})$, i.e. $H_{\text{Gram}}(V) \leq \sum_{i=1}^m H_{\text{Gram}}(\mathbf{v}_i)$ where $H_{\text{Gram}}(\mathbf{v}_i) \equiv \log(K_{ii})$. Equality holds iff off-diagonal inner products $K_{ij} = \langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0$ for all $i \neq j$. ■

We flag explicitly that H_{Gram} , as used in this theorem, shares the name "entropy" with H_{ling} (Eq. 5) but is a different mathematical object — a continuous log-determinant quantity, versus a bounded discrete Shannon entropy over normalized singular values. The subadditivity law of Eq. (10) should be understood as applying to H_{Gram} .

4. INFORMATION-GEOMETRIC RICCI FLOW & HARDWARE ACCELERATION

4.1 Emergent Geometry Engine

In PMCA Generation 6.0, geometry is no longer pre-selected. The manifold starts flat ($\mathbf{g}_{ij} = \mathbf{I}$) and self-structures entirely from data statistics via `fractal_undetermined_geometry()`.

4.2 Local Information-Geometric Metric Evolution and Stability Analysis

Rather than assuming a static spatial geometry, PMCA updates the point-wise Riemannian metric tensor $\mathbf{g} \in \mathcal{S}_{++}^3$ dynamically along continuous state updates τ , via a matrix differential equation *inspired by* Hamilton-Perelman Ricci flow [64], [65], combined with data-driven Fisher Information Metrics [66]:

$$\frac{dg_{ij}}{d\tau} = -2\mathcal{R}_{\text{norm}}g_{ij} + \alpha F_{ij}(\text{data}), \quad (11)$$

where F_{ij} represents the regularized Fisher Information matrix:

$$F_{ij} = (1 + H_{\text{spectral}}) \left(\delta_{ij} + \frac{d_i d_j}{\|\mathbf{d}\|^2 + \epsilon} \right) \quad (12)$$

with H_{spectral} representing normalized Shannon spectral entropy. To avoid computing continuous spatial derivatives across discrete state vectors on-chip, the intrinsic geometric deviation is evaluated via the Isotropic Metric Anisotropy Index $\mathcal{R}_{\text{norm}}$, an engineering **proxy** for scalar curvature — not the Ricci curvature tensor itself:

$$\mathcal{R}_{\text{norm}} \approx \frac{(\text{tr}(\mathbf{g}) - 3)^2 + 2(g_{12}^2 + g_{13}^2 + g_{23}^2)}{|\det(\mathbf{g})| + \epsilon}. \quad (13)$$

*[**AUTHOR ACTION NEEDED:** confirm whether the current implementation computes $\mathcal{R}_{\text{norm}}$ via Eq. (13), a genuine Ricci tensor contracted from the Christoffel connections of Section 5.2, or something else — and make sure this section's language matches whichever is actually running, so the manuscript's claim about what "Ricci-flow-inspired" means stays accurate to the implementation.]*

Proposition 4 (Strict Positive-Definiteness of F_{ij}).

For any non-zero data vector $\mathbf{d} \in \mathbb{R}^3$ and spectral entropy $H_{\text{spectral}} \geq 0$, the Fisher Information Metric matrix F_{ij} defined in (12) is strictly positive-definite ($F_{ij} \succ 0$).

Proof: For any non-zero vector $\mathbf{v} \in \mathbb{R}^3$, $\mathbf{v}^T \mathbf{F} \mathbf{v} = (1 + H_{\text{spectral}}) \left(\|\mathbf{v}\|^2 + \frac{(\mathbf{v}^T \mathbf{d})^2}{\|\mathbf{d}\|^2 + \epsilon} \right)$. Since $1 + H_{\text{spectral}} \geq 1.0$ and $(\mathbf{v}^T \mathbf{d})^2 \geq 0$, $\mathbf{v}^T \mathbf{F} \mathbf{v} \geq \|\mathbf{v}\|^2 > 0$, proving $\mathbf{F} \succ 0$. ■

Proposition 5 (Metric Stability under Flow Integration).

Let $\mathbf{g}^{(0)} = \mathbf{I} \succ 0$. For a discrete integration step size $\eta \leq \frac{1}{2 \max \|\mathcal{R}_{\text{norm}} \mathbf{g}\|_2 + \alpha \|F_{ij}\|_2}$, the updated metric tensor $\mathbf{g}^{(\tau+1)} = \mathbf{g}^{(\tau)} + \eta(-2\mathcal{R}_{\text{norm}} \mathbf{g}^{(\tau)} + \alpha F_{ij})$ remains within the open cone of Symmetric Positive-Definite (SPD) matrices $\mathcal{S}_{++}^3$.

Proof sketch: By Proposition 4, $F_{ij} \succeq \mathbf{I}$. By Weyl's inequality, the smallest eigenvalue of a perturbed matrix satisfies $\lambda_{\min}(\mathbf{g}^{(\tau)} + \Delta) \geq \lambda_{\min}(\mathbf{g}^{(\tau)}) - \|\Delta\|_2$ for any symmetric perturbation Δ . Setting $\Delta = \eta(-2\mathcal{R}_{\text{norm}} \mathbf{g}^{(\tau)} + \alpha F_{ij})$ and bounding η as stated ensures $\lambda_{\min}(\mathbf{g}^{(\tau+1)}) > 0$, preventing metric degeneration. ■

4.3 XLA Geometry Transcription Layer

To bridge unlimited mathematical geometry with XLA's fixed-shape tensor constraints, the transcription layer encodes arbitrary 3-manifold metric tensors into a shape-stable $(N, 6)$ array containing the 6 unique elements of the symmetric 3×3 matrix $[g_{11}, g_{12}, g_{13}, g_{22}, g_{23}, g_{33}]$. The shape never changes; the values evolve freely, which avoids XLA graph recompilation on Google TPU v5e accelerators.

4.4 Potential-First Principle and Convergence

Recursion continues down fractal depth until the metric stabilizes at an exact fixed point:

$$\|\Delta \mathbf{g}\|_2 < \epsilon \quad (\epsilon = 10^{-6}). \quad (14)$$

4.5 Geometric Information Capacity Index ($\mathcal{C}_{\text{geom}}$)

To evaluate structural state capacity without incurring the exponential combinatorial complexity $\mathcal{O}(2^N)$ of Tononi's Integrated Information Theory (Φ) [70], PMCA evaluates a scalar Geometric Information Capacity Index:

$$\mathcal{C}_{\text{geom}} = \text{mean}(\log(|\det(\mathbf{g})| + 1)) \times \text{mean}(\text{tr}(\mathbf{g})). \quad (15)$$

Geometrically, $\mathcal{C}_{\text{geom}}$ serves as an operational scalar indicator coupling logarithmic volume expansion with average metric scaling. It acts as an efficient $\mathcal{O}(1)$ on-device monitor for metric non-degeneracy and anisotropy during XLA execution, ***without asserting formal equivalence to axiomatic information-integration measures.*** $\mathcal{C}_{\text{geom}}$ (Eq. 15) is the only information-capacity-adjacent quantity formally defined in this manuscript.

[AUTHOR ACTION NEEDED:** confirm that no field or variable in the accompanying implementation is named in a way that invokes IIT's Φ (e.g., `integrated\_information\_Phi`) without corresponding to a quantity actually defined in this manuscript. If such a field exists, rename it to avoid implying an equivalence this section explicitly disclaims.]**

4.6 Integrity Verification Protocol

This protocol guarantees zero language probability or persona bias enters the geometry engine. Data flow is strictly one-way: geometry codec output is a read-only, immutable `FrozenGeometrySnapshot`. Cryptographic SHA-256 state hashing verifies that geometry states remain untouched during translation passes. We note this protocol verifies data **integrity** (tamper-evidence via hashing); it does not provide process-level isolation or sandboxing in the systems-security sense, and should not be conflated with such guarantees.

4.7 Geometry Anomaly Discovery Engine

Extracts a topological fingerprint S_{topo} capturing trace variation, volume element, scalar curvature, anisotropy, and shear balance. Compares fingerprints against reference families (Euclidean, Spherical, Hyperbolic, Toroidal, Anisotropic High-Dimensional). If no match exceeds the confidence threshold, the manifold is flagged as an out-of-distribution candidate novel geometry.

[AUTHOR ACTION NEEDED: this section needs the explicit formal definition of each fingerprint dimension and the confidence-threshold decision rule before submission — currently descriptive only, at a lower rigor level than the rest of Section 3–4.]

5. DUAL-ENGINE REASONING AND METACOGNITIVE VERIFICATION

5.1 Dual-Engine Operator Manifold

The `OperatorManifold` executes two parallel paths [17], [43]:

- **Path A (Analytical Symbolic Substrate):** Evaluates SymPy closed-form AST equations, Chebyshev Jacobian operator matrices, and SVD spectral entropy H_{math} [18].

- **Path B (Empirical Physical Substrate):** `PhysicsEngine3D` executes numerical 3D particle dynamics (Newtonian N-body gravitation, Hooke spring forces, drag, elastic collisions) using symplectic Euler integration ($v \leftarrow 0.99v$, $\Delta t = 0.016$), preserving long-horizon physical stability [44], [46].

[AUTHOR ACTION NEEDED: the manuscript should state explicitly what Path B contributes to verification of a typical symbolic query, beyond generating coordinates for the Section 6 3D visualization. If Path B's primary role is visualization/telemetry rather than independent verification, say so directly — that is a legitimate design, but "dual-engine... verification" currently implies a stronger coupling than is demonstrated. Separately: confirm whether the deployed system runs with a live LLM adapter attached, or with the adapter set to null as in the reference implementation — if the latter, the comparison framing in Section 1 and Section 6 should be scoped explicitly to "verification of closed-form math," not general LLM replacement.]

5.2 Riemannian RK4 Geodesic Integration

In the curved conal phase-space, state trajectories follow non-Euclidean geodesics governed by the Euler-Lagrange equations $\ddot{x}^k + \Gamma_{ij}^k \dot{x}^i \dot{x}^j = 0$. Because the metric tensor field $\mathbf{g}(z, r, \theta)$ is known analytically via Eq. (2), the Christoffel symbols of the second kind:

$$\Gamma_{ij}^k = \frac{1}{2} g^{kl} (\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij}) \quad (16)$$

are evaluated using exact analytical partial derivatives with respect to conal coordinates (z, r, θ) :

$$\frac{\partial g_{11}}{\partial z} = \frac{1}{Z_{\max}}, \quad \frac{\partial g_{22}}{\partial z} = -\frac{\alpha}{Z_{\max}}, \quad \frac{\partial g_{33}}{\partial \theta} = -0.1 \sin \theta, \quad (17)$$

with all cross-partial terms vanishing identically due to the diagonal structure of $\mathbf{g}(z, r, \theta)$. This eliminates numerical finite-difference discretization noise entirely, reducing trajectory integration error strictly to standard 4th-Order Runge-Kutta (RK4) truncation terms of $\mathcal{O}(\Delta t^4)$ [51]:

$$\mathbf{k}_1 = f(\mathbf{y}_n), \quad \mathbf{k}_2 = f\left(\mathbf{y}_n + \frac{\Delta t}{2} \mathbf{k}_1\right), \quad \dots \quad (18)$$

5.3 Metacognitive Dialectic Self-Interrogation

Prior to generating output representations through the translation pass, Phase 12 (`SelfInterrogationEngine`) executes formal dialectic verification by probing asymptotic limits of the state trajectory $\mathbf{v}(z)$ along longitudinal boundary points $z_k \in \{0, Z_{\max}\}$:

$$\text{Thesis } \mathbf{v}(z) \quad \longleftrightarrow \quad \text{Antithesis } \left(\lim_{z \rightarrow 0^+} \mathbf{v}(z), \lim_{z \rightarrow Z_{\max}^-} \mathbf{v}(z) \right) \quad \longrightarrow \quad \text{Synthesis.}$$

The engine evaluates domain continuity, divergence, and essential singularities. We define the normalized Dialectic Resilience Score $S_{\text{dialectic}} \in [0, 1]$ as:

$$S_{\text{dialectic}} = \mathbb{I}(\mathcal{D}_{\text{sing}} = \emptyset) \cdot \exp \left(-\beta \sum_{k \in \{0, Z_{\text{max}}\}} \left\| \lim_{z \rightarrow z_k} \mathbf{v}(z) - \mathbf{v}_{\text{boundary}} \right\|_2 \right), \quad (19)$$

where $\mathbb{I}(\cdot)$ is an indicator function returning 0 if an undefined singularity ($\mathcal{D}_{\text{sing}}$, e.g., NaN, ComplexInfinity) is detected, and 1 otherwise. The hyperparameter $\beta > 0$ controls boundary discontinuity penalization (set to $\beta = 1.0$ across all experiments). Trajectories receiving $S_{\text{dialectic}} < 1.0$ are rejected or routed for iterative metric smoothing prior to decoding.

6. EMPIRICAL EVALUATION AND BENCHMARK ANALYSIS

6.1 Experimental Setup

All empirical benchmarks were conducted using JAX/XLA hardware acceleration on Google Colab attached to a Cloud TPU v5e slice (`TpuDevice(id=0)`, native bfloat16/float32 vector units).

6.2 Live-Measured Telemetry and Verification

To validate the operational properties of the Dual-Engine Operator Manifold (§5.1), we evaluate symbolic calculus execution and physical field induction.

Across symbolic targets, the SymPy closed-form AST parser achieved exact mathematical verification without algebraic drift (*benchmarks to be completed*). Differential geometry execution demonstrated exact conservation of the fully-integrated simulated velocity field, with a measured divergence of $\text{div}(\mathbf{V}) = 0.00000000$, consistent with the antipodal force-vector cancellation established in Theorem 2 together with the composite dynamics of Path B (§5.1); see the Remark following Theorem 2 for the precise distinction between this simulated-field quantity and the closed-form casing field's (necessarily non-zero) local divergence.

Table 1 presents live-measured substrate metrics. Cold processing of calculus queries completes in 172.35 ms, while canonical SHA-256 hash lookup via `RelatableValueCategorizer` drops latency to 0.0350 ms (35 μ s), yielding an in-memory retrieval speedup of $4,924\times$.

Table 1: Factual Empirical Metrics Measured on PMCA Substrate (Cloud TPU v5e)

Measured Performance Dimension	Live Empirical Value
Cold Pipeline Execution Latency	172.35 ms
Zero-Reprocessing ($\mathcal{O}(1)$ Cache Hit) Latency	0.0350 ms (35 μ s)
Cache Retrieval Speedup Factor	4,924 $\times$
Proof-of-Concept Symbolic Verification Rate	(<i>benchmarks to be completed</i>)
Symbolic Math Hallucination Rate	0.00%
Long-Horizon Energy Preservation (1,000 steps)	94.31% ($T_0 = 7.50\text{J} \rightarrow T_1 = 7.07\text{J}$)
Simulated Net Velocity Field Divergence	0.00000000
Peak Device Geometry Buffer Footprint*	0.01 MB (10.2 KB)
Single-Pass 10M Particle Hot Execution Latency	21.73 ms
Sustained 10M Particle Single-Pass Throughput	460,184,324 particles/s

**Note on Memory Footprint:* The 0.01 MB (10.2 KB) figure measures the active XLA device-side tensor allocation for the shape-stable $(N, 6)$ metric array at the nominal fingerprint grid size $N = 216$ (18 topological channels $\times$ 12 feature dimensions). Total Python host runtime footprint (including JAX runtime binaries, SymPy AST parser, and Sentence-Transformer embedding weights) consumes 142.8 MB of host RAM.

6.3 Systematic Component Ablation Study

Table 2: Systematic Component Ablation Study

Substrate Configuration	Proof Acc. (%)	Simulated Field Divergence	Mean Latency (ms)
Full PMCA System	<i>(benchmarks to be completed)</i>	0.00000000	172.35
w/o Gradient Potential Drive	<i>(benchmarks to be completed)</i>	0.00412000	128.48
w/o Encasing Vector Casing	<i>(benchmarks to be completed)</i>	0.14280000	145.20
w/o Dialectic Interrogation	<i>(benchmarks to be completed)</i>	0.00000000	134.10
w/o SymPy AST Verification	<i>(benchmarks to be completed)</i>	0.00000000	98.60
w/o SHA-256 Cache (Cold Only)	<i>(benchmarks to be completed)</i>	0.00000000	172.35

6.4 Comparative Baseline Analysis

To isolate the verification efficacy of PMCA's Dual-Engine Operator Manifold, we evaluated symbolic verification performance against two standard baselines on **closed-form calculus tasks specifically**: (1) Direct SymPy CPU Execution (without conal metric trajectories or dialectic interrogation), and (2) Zero-Shot LLM Translation (GPT-4o prompting for symbolic calculus resolution). This comparison is scoped to the domain PMCA's SymPy-based Path A directly addresses, not to general-purpose language understanding.

Table 3: Comparative Symbolic Verification Performance Across Boundary Calculus Operations

Architecture	Verification Acc. (%)	Unhandled Exceptions (%)	Mean Latency (ms)
Zero-Shot LLM (GPT-4o)	<i>(benchmarks to be completed)</i>	0.0%	1,240.00
Direct SymPy (CPU)	<i>(benchmarks to be completed)</i>	14.0%	112.50
PMCA Substrate (Ours)	<i>(benchmarks to be completed)</i>	0.0%	172.35

6.5 Vectorized Particle Scaling and Chunking Analysis

When scaling metric transformations to extreme volume, monolithic tensor allocations incurred an HBM memory wall at 100,000,000 particles, causing execution latency to jump

to 50,637.75 ms due to intermediate activation thrashing during transcendental cosh/sinh evaluations [24]. Implementing a chunked XLA batching strategy ($N = 10^9$ particles divided into $K = 20$ chunks of $S = 50,000,000$ particles) eliminated memory thrashing by keeping intermediate activation buffers within fast on-chip HBM cache [62].

Table 4: Vectorized Particle Transformations on Cloud TPU v5e

Workload Scale (N)	Batch Strategy	Time (ms / s)	Throughput (particles/s)
100K particles	Monolithic JIT	1.43 ms	69.93×10^6
10M particles	Monolithic Cold	4,047.96 ms	2.47×10^6
10M particles	Monolithic Hot	3,702.83 ms	2.70×10^6
100M particles	Monolithic (HBM wall)	50,637.75 ms	1.97×10^6
1B particles	20 chunks $\times$ 50M	22,772.70 ms	43.91×10^6
100B particles	200 chunks $\times$ 500M	273.75 s	365.30×10^6

6.6 PPCTL Translation Pass Evaluation

[AUTHOR ACTION NEEDED: this subsection currently evaluates `TinyLanguageCodec` without ever defining it in body text (it appears only in Table 5's file mapping). Before submission, add a short formal description here — e.g., is it a deterministic template renderer, a rule-based grammar, or something else? Given every other pipeline stage in this manuscript gets an explicit equation or algorithm, this is the one black box left, and it sits exactly where Section 1 claims to solve "black-box opacity" — closing this gap directly strengthens that claim.]

Evaluating the Post-Processing Communicative Translation Layer on derived physical solutions confirms that natural language output decoding introduces a deterministic overhead of just 0.85 ms. The rendered natural language translation was evaluated against the underlying SymPy AST representation and confirmed to match its mathematical content, indicating no semantic drift across the PPCTL interface for the cases tested.

7. DISCUSSION, LIMITATIONS, AND FUTURE WORK

7.1 Architectural Implications

For the domain directly demonstrated — closed-form symbolic mathematics — PMCA Generation 6.0 shows that routing verification through a symbolic solver eliminates linguistic hallucination for that domain. By executing state transformations in a fixed, low-dimensional metric geometry and verifying mathematical invariants ($\mathcal{C}_{\text{geom}} > 0$), PMCA provides a deterministic, inspectable alternative to uninterpretable attention weights **for this domain**; we have not demonstrated this property for open-ended natural-language reasoning, and do not claim it does.

7.2 Limitations

While high-volume element-wise metric scaling parallelizes effectively on hardware accelerators, sequential symbolic algebra parsing (SymPy AST generation) remains CPU-bound, requiring 100 ms–500 ms per complex user query. The curvature term $\mathcal{R}_{\text{norm}}$ (Eq. 13) is an anisotropy-based proxy rather than a Ricci tensor computed from the manifold's own Christoffel connections (Eq. 16); replacing it with a fully-derived Ricci tensor is left to future work. The role of Path B (§5.1) in independently verifying, rather than visualizing, symbolic results is not yet formally established and is likewise left to future work. Future work will also focus on compiling symbolic AST parsers directly into native C++/CUDA/XLA primitives, and on formally specifying the PPCTL text-rendering layer (§6.6).

8. CONCLUSION

The Pure Mathematical Conal Architecture (PMCA Generation 6.0) establishes a deterministic neuro-symbolic substrate that provides exact, non-hallucinating verification for closed-form mathematical reasoning, backed by a fixed-geometry, inspectable state-tracking layer. Empirical benchmarks demonstrate chunked scaling to 100 billion particle transformations in 273.75 s (365.3×10^6 particles/s) and sub-millisecond state retrieval (0.0350 ms) on Cloud TPU v5e hardware. We have scoped our claims to the domains directly demonstrated, and identify the extension of these properties to open-ended natural-language reasoning as an open problem for future work, not a solved one.

APPENDIX

Table 5: Cross-Reference Mapping of Theoretical Substrates to Codebase Implementation Files

Section	Mathematical Concept	Source File	Primary Class / Function
§3.1	Conal metric $\mathbf{g}(z, r, \theta)$ (Eq. 2)	<code>conal_tensor_pathway.py</code>	<code>ConalTensorPathway.process_core</code>
§3.1	XLA JIT metric scaling	<code>jax_conal_pathway.py</code>	<code>@jax.jit jax_conal_metric_scaling</code>
§3.2	Casing force $\mathbf{F}_{\text{ext}}$ (Eq. 3)	<code>external_conal_casing.py</code>	<code>ExternalVectorStructure.compute</code>
§3.3	Dense semantic embedding	<code>next_gen_stack.py</code>	<code>DenseTransformerEmbedder.embed</code>
§3.4	Potential field $V(z)$ (Eq. 8)	<code>gradient_potential.py</code>	<code>GradientPotentialDrive.compute</code>
§3.5	Entropy subadditivity (Eq. 10)	<code>entropy_dissipation_law.py</code>	<code>PMCAConalEntropyDissipationLaw</code>
§4.1	Emergent Geometry Engine	<code>jax_fractal_undetermined_geometry.py</code>	<code>fractal_undetermined_geometry</code>
§4.2	Information Ricci Flow (Eq. 11)	<code>jax_fractal_undetermined_geometry.py</code>	<code>@jax.jit ricci_flow_step()</code>
§4.3	Shape-Stable $(N, 6)$ Packing	<code>xla_geometry_transcription.py</code>	<code>@jax.jit transcribe_metric_to_array</code>
§4.4	Potential-First Convergence	<code>jax_fractal_undetermined_geometry.py</code>	<code>fractal_recursive_evolution()</code>
§4.5	Geometric Info Capacity $(\mathcal{C}_{\text{geom}})$	<code>jax_fractal_undetermined_geometry.py</code>	<code>compute_cgeom()</code>
§4.6	Integrity Verification Protocol	<code>formal_isolation_protocol.py</code>	<code>IsolationProtocol.export_frozen_snapsh</code>
§4.7	Geometry Anomaly	<code>geometry_discovery_engine.py</code>	<code>@jax.jit classify_topology()</code>

Section	Mathematical Concept	Source File	Primary Class / Function
	Discovery Engine		
§5.1	SymPy AST calculus solver	<code>mathematical_substrate.py</code>	<code>MathematicalSubstrate.evaluate</code>
§5.1	3D particle classical physics	<code>physics_engine_3d.py</code>	<code>PhysicsEngine3D.step()</code>
§5.2	Christoffel symbols Γ_{ij}^k (Eq. 16)	<code>riemannian_diff_geometry.py</code>	<code>ChristoffelConnectionCalculator</code>
§5.2	RK4 geodesic integrator (Eq. 17)	<code>riemannian_diff_geometry.py</code>	<code>GeodesicIntegrator.integrate_curve</code>
§5.3	Dialectic self-interrogation (Eq. 18)	<code>self_interrogation_engine.py</code>	<code>SelfInterrogationEngine.execute</code>
§6.2	SHA-256 zero-reprocessing cache	<code>relatable_value_categorizer.py</code>	<code>RelatableValueCategorizer.lookup</code>
§6.6	Tiny Language Codec (PPCTL)	<code>tiny_language_codec.py</code>	<code>TinyEncoder.encode()</code>

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During preparation of this manuscript, the author used Antigravity (Google DeepMind) and Claude (Anthropic) as AI-assisted tools for software refactoring support, mathematical derivation checking, and manuscript formatting, under the author's direct supervision. All mathematical claims, proofs, and experimental results in this manuscript were reviewed by the human author, who bears sole responsibility for their accuracy. No AI system is credited as an author or contributor to this work.

[AUTHOR ACTION NEEDED: confirm the target venue before final submission, since abstract length limits, citation style, and AI-disclosure placement conventions vary by publisher — the wording above follows the general disclosure-not-authorship norm shared

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